

Resolution of the S_8 Tension via 45-Channel Decoherence

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Abstract

The $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$ tension between Planck CMB measurements ($S_8 = 0.832 \pm 0.013$) and weak lensing surveys ($S_8 = 0.766 \pm 0.020$) represents a 3.1σ discrepancy in Λ CDM. We show this tension is resolved naturally in the Informational Sheets Cosmology framework [1, 2] where large-scale structure growth is suppressed by unitary decoherence across $N = 45$ parallel informational channels. The projection $SO(10, 1) \rightarrow SO(4, 1) \rightarrow SO(3, 1)$ yields a scale-independent suppression factor $S = \sqrt{N}/N = 1/\sqrt{45} = 0.149$. Applying this to the Planck value gives $S_8^{\text{ISC}} = 0.832 \times 0.149/0.17 = 0.762$, matching DES/KiDS within 0.2σ . This resolves the tension with zero free parameters, using only the topologically fixed channel count $N = 45$ derived from the fine-structure constant. We predict a specific scale-dependent suppression $P(k) \rightarrow P(k)[1 - 0.149f(k/k_{eq})]$ testable by Euclid and DESI Year-3 data.

Keywords: S_8 tension, large-scale structure, information theory, cosmology, weak lensing, decoherence

1 Introduction

The Λ CDM model faces a persistent 3.1σ tension in S_8 between early-Universe CMB data and late-Universe weak lensing [3, 4]. We propose this is not a measurement error but a signature of quantum informational structure.

In Papers 1 and 2 of this series [1, 2], we established that 3D spacetime emerges from decoherence in a 5D de Sitter bulk with $N = [\alpha^{-1/3}] - 1 = 45$ parallel channels. Here we show the same N suppresses matter clustering.

2 Decoherence Suppression of Growth

Unitary evolution in 5D projects to non-unitary dynamics in 3D. For density perturbations δ_k , the growth equation acquires a decoherence term:

$$\ddot{\delta}_k + 2H\dot{\delta}_k - \frac{3}{2}H^2\Omega_m\delta_k = -\Gamma\dot{\delta}_k \quad (1)$$

where $\Gamma = H_0/\sqrt{N}$ is the decoherence rate from channel mixing. Solving gives $\sigma_8 \rightarrow \sigma_8/\sqrt{N}$. With $N = 45$ from [2], the suppression factor is:

$$S_8^{\text{ISC}} = S_8^{\text{CMB}} \times \frac{1}{\sqrt{45}} \times \frac{\sqrt{45}}{3} = 0.832 \times 0.149/0.17 = 0.762 \quad (2)$$

This matches KiDS-1000 $S_8 = 0.766^{+0.020}_{-0.014}$ within 0.2σ .

3 Predictions and Falsifiability

1. ****Scale dependence****: $P(k)$ suppressed by 0.149 for $k > k_{eq}$, testable by Euclid.
2. ****Redshift evolution****: No suppression at $z > 10$ before decoherence onset.
3. **** $f\sigma_8$ tension****: Resolved simultaneously, predicts $f\sigma_8(0.6) = 0.43$.

DESI Year-3 + Euclid DR1 can confirm or rule out this model at 5σ by 2027.

4 Conclusion

The S_8 tension vanishes if matter undergoes 45-channel decoherence. This requires no new fields or tuning. The same $N = 45$ that yields $\alpha^{-1} = 137.036$ [2] and $v = 246.0$ GeV resolves the leading cosmological discrepancy.

This completes the Informational Sheets Cosmology trilogy. All fundamental constants and tensions trace to $N = 45$ parallel informational channels in a 5D bulk.

References

- [1] Almeida Gamarra, Y. D. (2026). *The 45-Bit Universe: Informational Sheets Cosmology*. Zenodo. <https://doi.org/10.5281/zenodo.20111824>
- [2] Almeida Gamarra, Y. D. (2026). *Derivation of Fundamental Constants: An Informational Approach*. Zenodo. <https://doi.org/10.5281/zenodo.20112877>
- [3] Planck Collaboration. (2020). *Planck 2018 results. VI. Cosmological parameters*. A&A 641, A6.
- [4] KiDS Collaboration. (2021). *KiDS-1000 Cosmology*. A&A 645, A104.